

10.5 The global patterns of energy supply and demand

This chapter will explain:

- ★ the increasing global production and consumption of energy
- ★ trends in the consumption of energy
- ★ energy surplus and deficit and the importance of energy security
- ★ the reasons for variations in the types of energy produced and used.

The increasing global production and consumption of energy

The 2024 Statistical Review of World Energy noted that 2023 was a record year for:

- » the consumption of fossil fuels – global consumption of crude oil broke through the 100 million barrels per day level for the first time ever
- » global GHG emissions from energy – global energy-related GHG emissions exceeded 40 gigatonnes for the first time ever
- » the generation of electricity from renewable sources, driven by increasingly competitive wind and solar energy – the renewables share of total primary energy reached 14.6 per cent, while fossil fuel consumption as a proportion of primary energy continued to fall to 81.5 per cent.

The demand for and production of energy has changed over time due to the following factors.

Population growth

World population was 3 billion in 1960. It doubled to 6 billion in 1999, a phenomenal increase in just under 40 years. It then reached 8 billion in late 2022. Such a large population increase has created huge extra demand for energy in itself, without considering any other factors. Add to this:

- » the factors such as increasing wealth that are pushing up the demand for energy
- » the considerable variations in the rate of population growth around the world, placing different national and regional pressures on energy production and demand.

Increased wealth

There is a strong correlation between per capita national income and the consumption of energy. However, this relationship is not perfect as other factors, climate in particular, affect the demand for energy. Generally, as average incomes increase, living standards improve. This involves the increasing use of energy and the use of a greater variety of energy sources. For example, when they can afford to do so people are likely to buy refrigerators, washing machines, dishwashers and cars. Demand will increase in all sectors of the economy as average incomes continue to increase.

Technological advances

Technological advances involve both technical improvements in the energy sector itself and technological developments in the rest of the economy that affect energy production and demand. For example, in the energy sector:

- » nuclear electricity has only been available since 1954
- » oil and gas can now be extracted from much deeper waters than in the past
- » renewable energy technology is advancing at an incredible speed. The costs of onshore wind and solar power fell by around 40 per cent and 55 per cent respectively between 2016 and 2020.

In terms of the wider economy, new technology can change the nature of energy demand in a big way. At one time, most of the world's trains were powered by coal. Today, more trains around the world are either electric or diesel. In recent years, the production of electrically powered cars has increased rapidly, replacing petrol and diesel models. As the world moves to a low-carbon economy, new technologies and digitisation are creating huge changes in the way energy is created and used.

Changes in price

The relative prices of different types of energy can influence demand. Electricity production in the UK began switching from coal to gas over 25 years ago. Initially, the main reason was that power stations were cheaper to run on natural gas. At the time, the environmental gain was a secondary factor.

Environmental factors and public opinion

Public opinion can influence decisions made by governments. People today are much better informed about the environmental impact of energy resources and other economic activities than they were in the past.

There is now more comprehensive legislation regarding the development of major energy projects, including detailed **environmental impact assessments**. However, the requirements and their enforcement vary from country to country. Canada is an example of a country that has comprehensive environmental impact assessments.

The impact of Covid-19

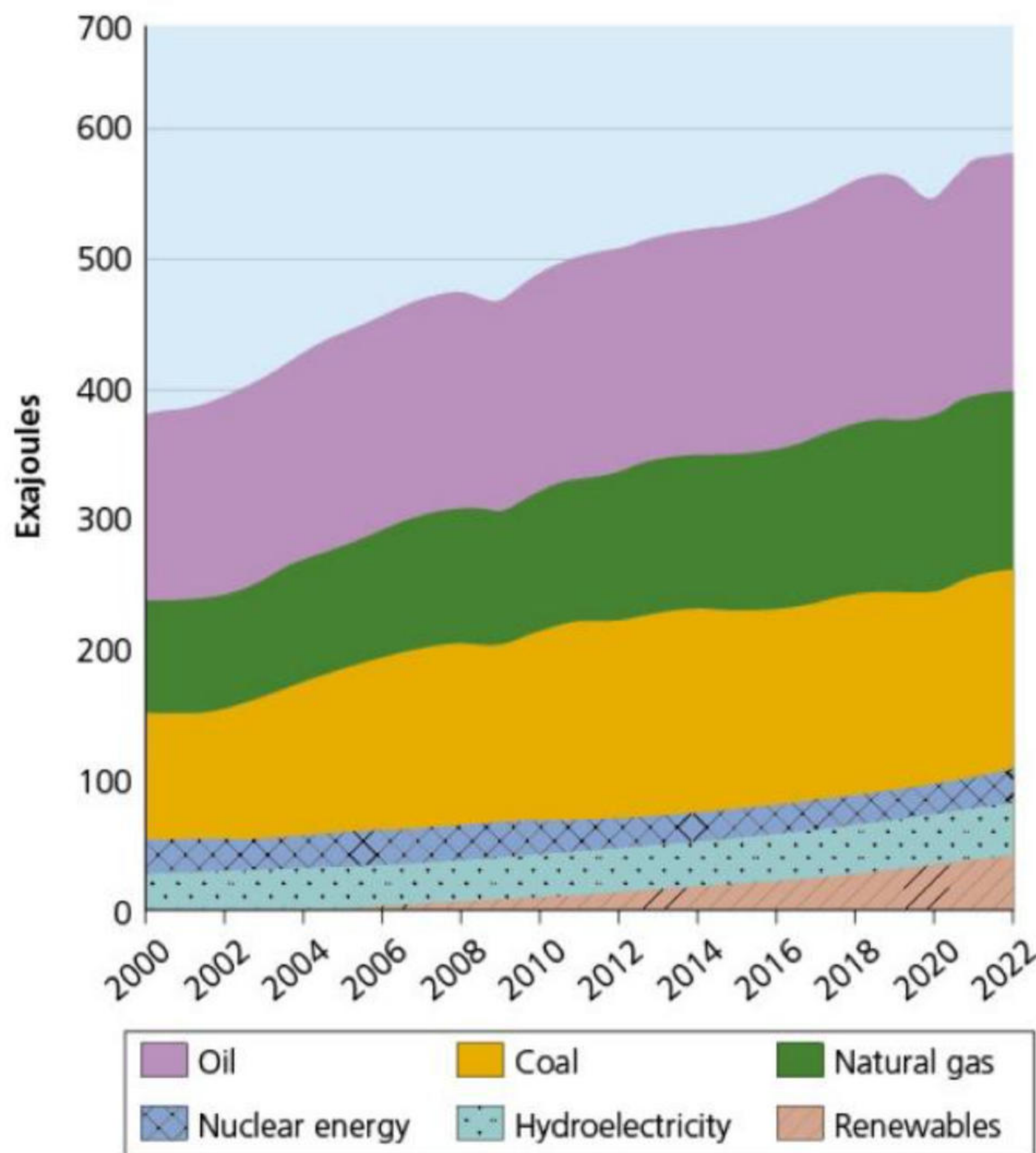
During the Covid-19 pandemic, the imposition of ‘lockdowns’ around the world had an astonishing impact on energy demand. The BP Statistical Review of World Energy 2021 described it as ‘one of the most tumultuous years for global energy in modern history’. The drop in energy consumption was mainly due to the sudden downturn in economic activity, which decimated transport-related demand. This led to falling demand for oil, which accounted for almost three-quarters of the decline. Overall, global energy demand fell by 4.5 per cent in 2020, the largest decline since the end of the Second World War. Carbon emissions from energy use fell by 6.3 per cent, to their lowest level since 2011. However, by 2023 the demand for energy had not only returned to pre-pandemic levels, but was beginning to exceed it.

Activity

Describe and explain two factors that have affected the demand for and production of energy.

Trends in the consumption of energy

Global energy consumption ([Figure 10.28](#)) increased by about 50 per cent between 2000 and 2023. Fossil fuels still dominate the global energy mix but their contribution is falling. Their relative contribution to primary energy consumption in 2023 was: oil 31.7 per cent, coal 26.5 per cent and natural gas 23.3 per cent. Together, the three fossil fuels accounted for 81.5 per cent of global primary energy consumption, down from 84.7 per cent in 2018. In contrast, hydroelectricity accounted for 6.4 per cent, nuclear energy 4.0 per cent and renewable energy 8.2 per cent. The latter was up from 4.1 per cent in 2018. Even in just a five-year time period, significant changes can occur.

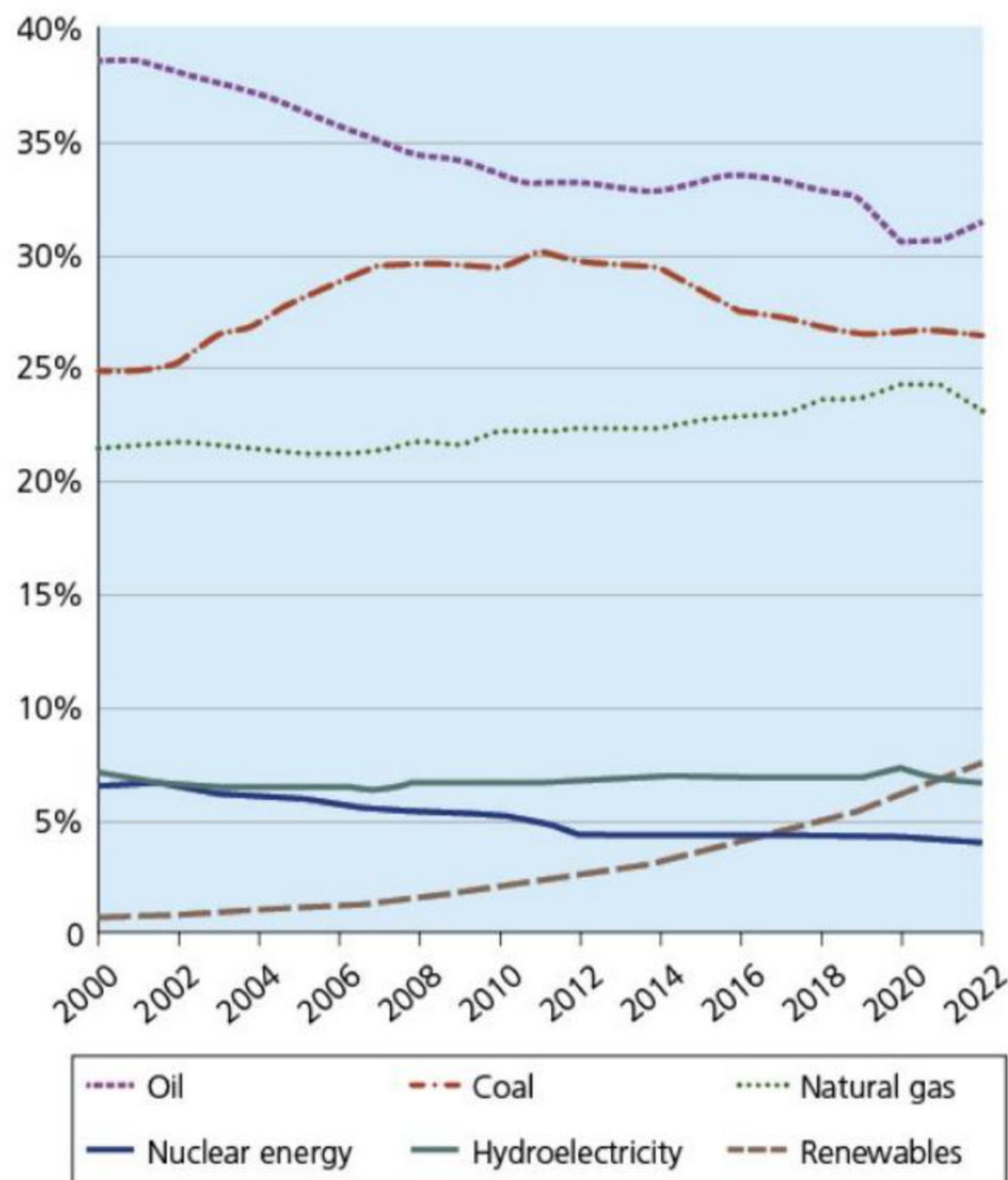


▲ **Figure 10.28** Global consumption of primary energy by source, 2000–22

[Figure 10.29](#) provides more detail about the changing contributions of the different sources of global primary energy. The declining share of oil is very clear, as is the rapidly rising contribution of renewable energy. Despite renewed interest in further development of nuclear energy, the declining global contribution of this source of energy is clear.

[Figure 10.30](#) shows the regional pattern of energy consumption for 2022. Consumption by type of fuel varies widely by world region. [Table 10.6](#) highlights some of the major contrasts between world regions. In 2023, the leading countries in total primary energy consumption were China, the USA, India, Russia and

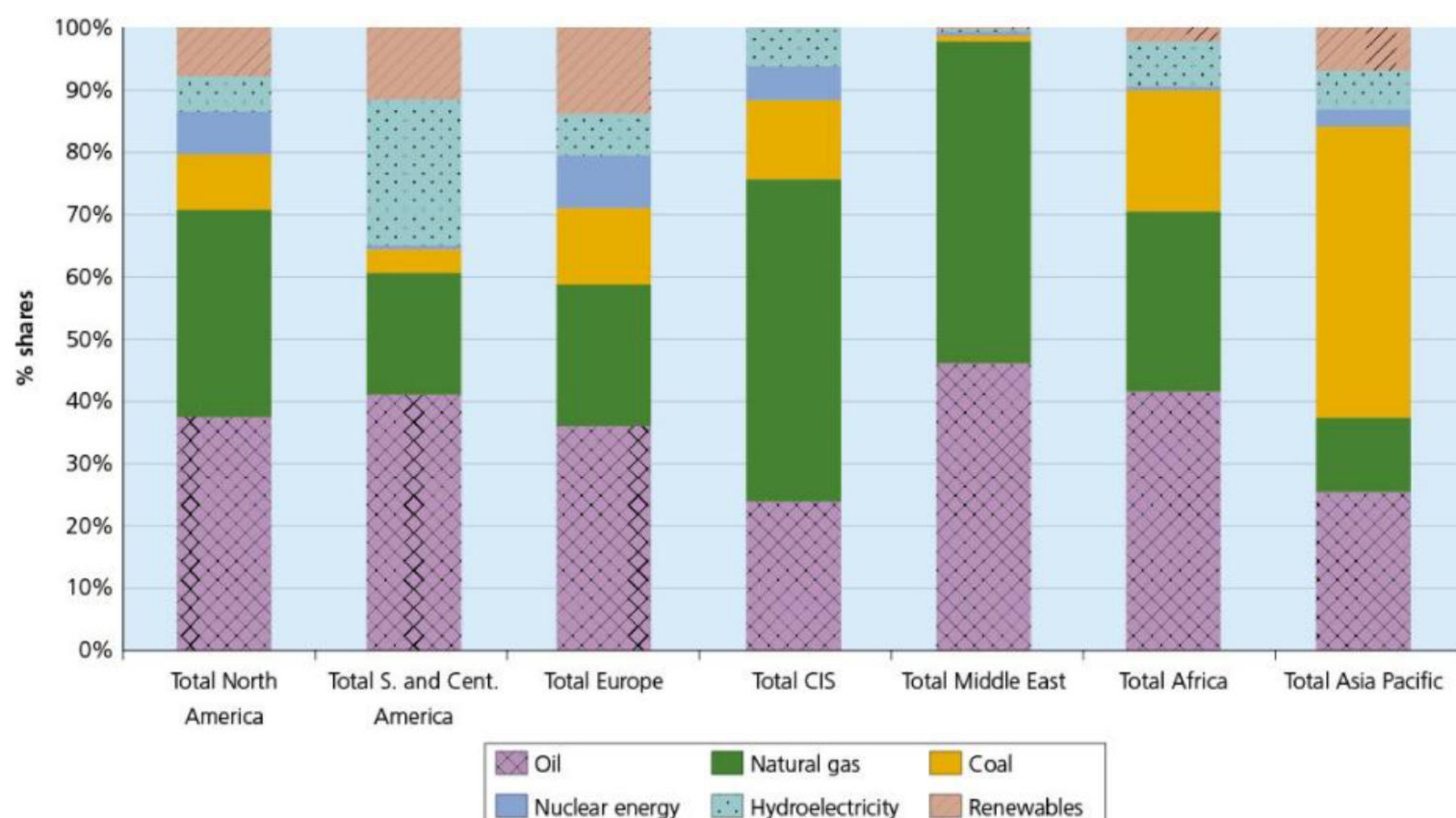
Japan. In terms of global share, the relevant figures were: China 27.6 per cent, the USA 15.2 per cent, India 6.3 per cent, Russia 5.1 per cent and Japan 2.8 per cent.



▲ **Figure 10.29** The changing share of global primary energy by source, 2000–22

▼ **Table 10.6** Major contrasts in the global energy mix

Energy source	Contribution to global energy mix in 2023
Oil	Only in the CIS (a grouping of Russia and eight other former Soviet Union countries) and Asia Pacific regions is the contribution of oil to total energy consumption less than 30 per cent. It is the main source of energy in four of the seven regions shown in Figure 10.30 . Oil makes its largest contribution to the energy mix in the Middle East (45.2 per cent)
Natural gas	Natural gas is the main source of energy in the Middle East (51.4 per cent) and in the CIS (52.7 per cent). Its lowest share of the energy mix is in Asia Pacific (11.3 per cent)
Coal	Only in the Asia Pacific region is coal the main source of energy. In contrast, it accounts for only 3.7 per cent of consumption in South and Central America, and 0.9 per cent in the Middle East. China was responsible for 56 per cent of global coal consumption in 2023
Nuclear energy	The highest relative importance of nuclear energy is in Europe (8.5 per cent), Northern America (7.0 per cent) and the CIS (5.1 per cent)
Hydroelectricity	The relative importance of hydroelectricity is greatest in South and Central America, where it accounts for 22.4 per cent of the energy mix. Elsewhere its contribution varies from 7.7 per cent in Europe to less than 1 per cent in the Middle East
Renewable energy	Consumption of renewable energy other than hydroelectric power is rising rapidly, but from a low base. Renewable energy makes the largest contribution to energy consumption in Europe (15.2 per cent) and in South and Central America (13.0 per cent)



▲ **Figure 10.30** Regional consumption of fuel by type, 2022

[Table 10.7](#) shows per capita energy consumption by world region in 2013 and 2023. In 2023, average consumption in Northern America was 16 times that of Africa. However, on an individual country basis there are a number of countries that have much higher consumption figures. [Table 10.8](#) shows all countries that have a per capita consumption over 300 gigajoules.

▼ **Table 10.7** Primary energy consumption per capita, 2013 and 2023

World region	Gigajoules (2013)	Gigajoules (2023)
World	74.1	77.0
North America	244.0	230.0
South and Central America	59.5	58.3
Europe	130.2	115.2
CIS	155.8	163.7
Middle East	136.8	142.9
Africa	15.2	14.3
Asia Pacific	54.9	67.3

▼ **Table 10.8** Countries with the largest energy consumption per capita – over 300 gigajoules per capita

Country	Per capita consumption (2023)
Qatar	816.7
Singapore	577.0
UAE	539.4
Trinidad and Tobago	384.3
Kuwait	365.9
Norway	363.7
Canada	359.7
Oman	333.4
Saudi Arabia	313.9

Activities

- 1 Describe the changes in world energy consumption shown in [Figure 10.29 \(page 274\)](#).
- 2 To what extent do the types of energy consumption vary by world region?
- 3 Summarise the differences and trends in [Table 10.7](#).
- 4 Suggest reasons for the very high levels of energy consumption in the countries listed in [Table 10.8](#).

Energy surplus and deficit and the importance of energy security

An increasing number of countries are facing an '**energy gap**'. This is the difference between a country's rising demand for energy and its ability to satisfy this demand from its own resources. The energy gap is often greatest in countries that, over time, have seen demand steadily increase and domestic production of energy stabilise or decline. Major improvements in the international transportation of energy over the last 50 years or so (pipelines, grid connectors, liquefied natural gas tankers) have made the importation of energy easier and cheaper. To an extent, however, these improvements in energy infrastructure may have deterred some countries from investing in the development of their own potential sources of energy. In 2023, the total international trade in oil, gas and coal was 53 per cent higher than it was in 2000.

- » The Middle East is crucial to the functioning of the global energy market, although it has lost some of its dominance in recent decades. In 2022, five of the top ten oil producers and three of the top twenty natural gas producers were located in the region.
- » Since the 1980s, Europe has consistently been a net importer of energy. The biggest energy deficit for Europe is oil. In 2023, Europe's oil production met only 23 per cent of its demand.
- » The Asia Pacific region has been a net importer of energy since the 1980s. In 2023, its largest deficit was in oil as production met only 19 per cent of consumption.
- » Northern America's energy system has changed significantly in recent decades, with rapid growth in the production of unconventional oil and gas beginning in the early 2000s. As a result, this region changed from being a net importer of energy to a net exporter. In 2023, oil production in Northern America was 16 per cent higher than its domestic consumption.
- » Latin America and the Caribbean is a net exporter of crude oil and coal, but a net importer of oil products and natural gas.

A secure and reliable energy supply is vital for the efficient functioning of a country's economy and society. Many countries that have to import a significant amount of the energy they use have become very concerned about energy security. Key examples include Germany and Japan. The International Energy Agency (IEA) defines **energy security** as the 'uninterrupted availability of energy sources at an affordable price'. Countries that have a high **import dependency** are actively looking at ways to increase their domestic production of energy.

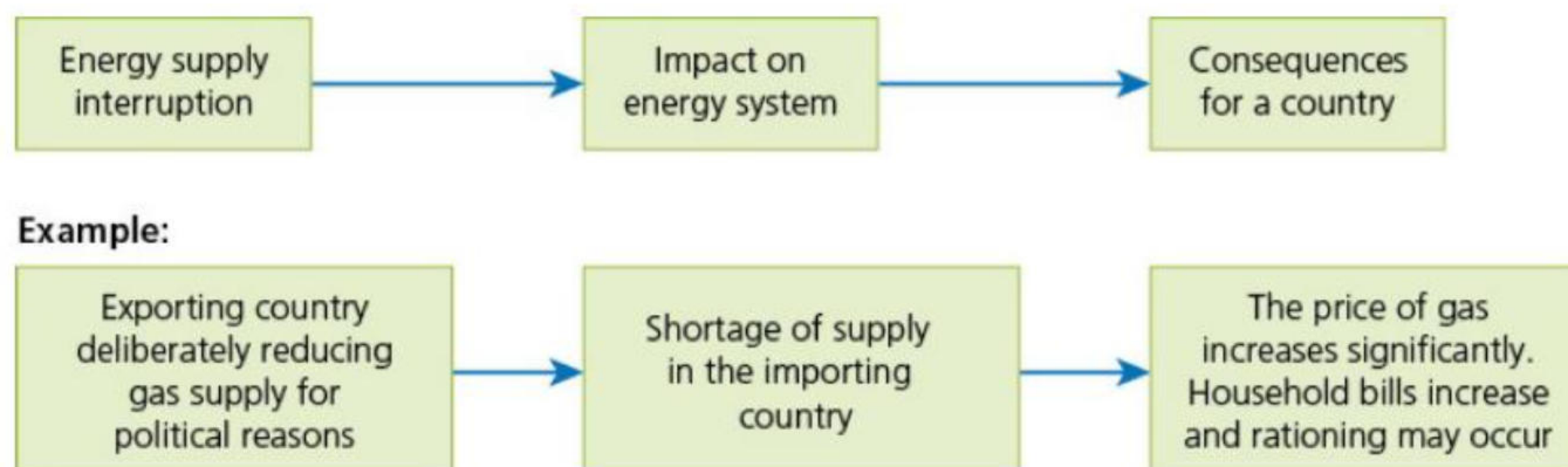
Energy security relates to present and future energy requirements:

- » Short-term energy security depends on the ability of an energy system to react quickly to sudden changes in the supply-demand balance.
- » Long-term energy security requires careful planning and significant investment in line with economic development and energy needs.

Energy supply interruption can be caused by a number of factors, including:

- » natural hazards, such as tropical storms and earthquakes
- » civil internal disturbance and international war
- » geopolitics, such as Russia reducing gas supply to European countries in 2022
- » rapid resource depletion, for example oil and gas fields running out
- » affordability
- » environmental concerns.

The energy supply interruption itself is the first stage in a three-stage sequence ([Figure 10.31](#)). The next stage is the impact of the interruption on an energy system. The final stage is the consequences resulting from such an impact. Consequences can vary from manageable to catastrophic.



▲ **Figure 10.31** The three stages of energy supply interruption

Countries with surplus energy can generally be considered to be energy secure. Countries including Canada and Russia earn valuable foreign currency by exporting energy to countries with limited domestic energy supplies. Where countries become over-reliant on energy from other countries, there is the risk that energy supply might be used as a political weapon. Africa is the world’s most energy-insecure continent. Lack of investment has resulted in insufficient power plant capacity, leading to low energy consumption and low access to power. Households connected to an energy grid often experience interruptions to supply.

Energy sources are affected by several factors including global climate change, population growth and economic development. In particular, extreme weather events such as drought, floods and rising temperatures are increasing demand for energy but also affecting supplies. Energy resources need to be safeguarded in order to meet the increased demands of population growth and economic development.

Energy supplies also depend on water. This is not just for hydroelectric schemes and biofuels but also for the production of fossil fuels and operating power plants. In addition, energy systems have become more vulnerable to cybercrime. Indeed, the energy sector is one of the UK’s top targets for cybercrime – nearly one-quarter of all cyberattacks in the UK were in the energy sector. Without doubt, energy supply has become a prominent factor in **geopolitics**.

As there is little excess capacity to ease pressure on energy resources, energy insecurity is increasing. The USA is seriously concerned about the political leverage associated with importing oil: in 1977 it began the construction of a **strategic petroleum reserve**. The oil is stored in huge underground tanks in southern Louisiana and Texas, and could easily be linked up to pipelines and shipping routes if needed. Since then, many other countries have built up oil reserves as protection against future energy crises. The USA’s strategic petroleum reserve was tapped after both Hurricane Katrina (2005) and Hurricane Gustav (2008) severely disrupted oil production in the US states bordering the Gulf of Mexico. Some countries also store reserves of coal and gas.

In recent decades concerns have developed about the security of key **energy pathways**, which might be blocked or destroyed as a result of political conflict. Most Middle East oil exports go by tanker through the Strait of Hormuz, a narrow body of water (55 km wide at its narrowest) between the Persian Gulf and the Gulf of Oman. Some countries have suggested that they might block the shipping route to prevent the oil getting through.

The reasons for variations in the types of energy produced and used

The reasons for variations in energy production can be subdivided into physical, economic and political factors. [Table 10.9](#) shows examples for each of these groupings. The combination of factors operating in each country can vary considerably. Some countries are rich in domestic energy resources, while others are energy resource poor. The latter countries rely heavily on imports of energy. However, energy resources by themselves do not constitute production. Capital and technology are also required to exploit resources.

▼ **Table 10.9** Factors affecting the production of energy

Physical	Economic	Political
● Deposits of fossil fuels are found in only a limited number of locations ● Large-scale hydroelectric power (HEP) development requires high precipitation, major steep-sided valleys and impermeable rock ● Large power stations require flat land and geologically stable foundations ● Solar power needs a large number of days each year with strong sunlight ● Wind power needs high average wind speeds throughout the year ● Tidal power stations require a very large tidal range ● The availability of biomass varies widely according to climatic conditions	● The most accessible, and lowest cost, deposits of fossil fuels are invariably developed first ● Onshore deposits of oil and gas are usually cheaper to develop than offshore deposits ● Potential HEP sites close to major transport routes and existing electricity transmission corridors are more economical to build than those in very inaccessible locations ● In LICs, FDI is often essential for the development of energy resources ● When energy prices rise significantly, companies increase spending on exploration and development	● Countries wanting to develop nuclear electricity require permission from the International Atomic Energy Agency ● International agreements such as the Kyoto Protocol can have a considerable influence on the energy decisions of individual countries ● Potential HEP schemes on ‘international rivers’ may require the agreement of other countries that share the river ● Governments may insist on energy companies producing a certain proportion of their energy from renewable sources ● Legislation regarding emissions from power stations favours the use of, for example, low-sulphur coal, as opposed to coal with a high sulphur content

A country’s **energy policy** can have a significant impact on demand if it focuses on efficiency and

sustainability as opposed to concentrating on building more power stations and refining facilities. High levels of pollution can be a strong stimulus to develop a cleaner energy policy.

Developing countries face many energy challenges, including:

- » poor security of supply
- » insufficient generating capacity
- » underdeveloped or non-existent grid infrastructure
- » a lack of adequate monitoring and control equipment and skilled human resources.

The use of fuelwood as a source of energy is largely confined to low- and lower-middle-income countries. In terms of other sources of energy, countries at this stage of development will largely rely on any domestic resources of fossil fuels if they have been able to gain access to the often high levels of investment required to develop such resources. Limited access to investment also goes a long way to explain the relatively low level of development of renewable energy sources in most low- and lower-middle-income countries. If domestic energy production is limited, significant levels of imports may be required. The types of fuel imported will be dictated by existing energy infrastructure and by differences in price.

Upper-middle-income countries have generally expanded their energy infrastructure in line with economic development and their increasing demand for energy. Typically, countries in this income bracket obtain energy from a wider range of non-renewable sources and will also have a growing renewable energy sector.

High-income countries have had the longest timespan in which to develop their energy infrastructures and the greatest availability of capital to adapt to changing challenges and opportunities. Such countries have generally invested the most in renewable energy and in commercial nuclear power. High levels of affluence allow for significant imports of energy if necessary and the ability to construct energy connections with neighbouring countries.

Activities

- 1 Explain the terms:
 - a Energy gap
 - b Energy security.
 - 2 Write a summary of [Figure 10.31 \(page 277\)](#).
 - 3 Why do many countries hold reserves of energy resources?
 - 4 What are energy pathways and why is their security so important?
 - 5 With reference to [Table 10.9](#), examine the physical factors that can affect the production of energy in a country.
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